

Ex. No. 3	ADD INTERACTIVITY TO APP – INPUT AND FORMS
Date of Exercise	
GitHub Host Link	https://github.com/Urk23cs8005/flut-exp3.git

Aim

To create an interactive Flutter student registration form with input fields, form validation, and a submit button.

Description

This experiment demonstrates how to create an interactive form in Flutter using `Form`, `TextFormField`, `GlobalKey<FormState>`, and `ElevatedButton`. The form accepts student name, email, and password. Validation messages are displayed when required fields are left empty.

Procedure:

1. Create a new Flutter project using the command:

```
flutter create student_form
```

2. Open the project folder:

```
cd student_form  
code.
```

3. Open the lib/main.dart file.
4. Create a `GlobalKey<FormState>` to manage form validation.
5. Create a `Form` widget and assign the form key.
6. Add a `TextFormField` for entering the student name.
7. Add an email input field using `TextInputType.emailAddress`.
8. Add a password field using `obscureText: true`.
9. Add an `ElevatedButton` for submitting the form.
10. Use the `validate()` method to check the entered information.
11. Display a success message using `SnackBar` when the form is valid.
12. Run the application and test the validation by clicking Submit without entering any information.

Program

```
import 'package:flutter/material.dart';

void main() {
  runApp(const StudentFormApp());
}

class StudentFormApp extends StatelessWidget {
  const StudentFormApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: const StudentForm(),
    );
  }
}

class StudentForm extends StatefulWidget {
  const StudentForm({super.key});

  @override
  State<StudentForm> createState() => _StudentFormState();
}

class _StudentFormState extends State<StudentForm> {
  final _formKey = GlobalKey<FormState>();

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Student Registration'),
      ),
      body: Padding(
        padding: const EdgeInsets.all(20),
        child: Form(
          key: _formKey,
          child: Column(
            children: [
              TextFormField(
                decoration: const InputDecoration(
                  labelText: 'Student Name',
                  border: OutlineInputBorder(),
                ),
              ),
            ],
          ),
        ),
      ),
    );
  }
}
```

```
        validator: (value) {
          if (value == null || value.isEmpty) {
            return 'Enter student name';
          }
          return null;
        },
      ),

      const SizedBox(height: 15),

      TextFormField(
        decoration: const InputDecoration(
          labelText: 'Email',
          border: OutlineInputBorder(),
        ),
        keyboardType: TextInputType.emailAddress,
        validator: (value) {
          if (value == null || value.isEmpty) {
            return 'Enter email';
          }
          return null;
        },
      ),

      const SizedBox(height: 15),

      TextFormField(
        obscureText: true,
        decoration: const InputDecoration(
          labelText: 'Password',
          border: OutlineInputBorder(),
        ),
        validator: (value) {
          if (value == null || value.isEmpty) {
            return 'Enter password';
          }
          return null;
        },
      ),
    ),
```

```
const SizedBox(height: 20),

ElevatedButton(
  onPressed: () {
    if (_formKey.currentState!.validate()) {
      ScaffoldMessenger.of(context).showSnackBar(
        const SnackBar(
          content: Text('Form submitted successfully'),
        ),
      );
    }
  },
  child: const Text('Submit'),
),
],
),
),
),
);
}
```

Output

The screenshot displays a mobile application interface titled "Student Registration Form". At the top, a purple header bar contains the title. Below the header, a light purple card shows a user profile with a graduation cap icon, the name "David", and registration details: "Reg No: URK23CS8005 | Course Code: 24CS2016". The main section is titled "Experiment 3: Form Inputs & Validation" and contains three input fields: "Student Name" (with a person icon and the text "David Paul"), "Email" (with an envelope icon and the text "davidpaul23@karunya.edu.in"), and "Password" (with a lock icon and masked text "*****"). A large purple "Submit" button is positioned below the fields. The application is running on an emulator, with a status bar at the bottom showing "01 September 2026" and "Tue 15:06 (Local time)". The system tray at the very bottom indicates "Dart 3.13.2 • Flutter 3.47.2" and shows various system icons including network, battery, and time.

Result

Thus, an interactive Student Registration Form was successfully created using Flutter. The application contains Name, Email, Password, Validation, and Submit Button, and displays appropriate validation messages for empty fields.